

How Many Cows Is a Life?

An honest attempt at the supply arithmetic of bovine oxygen carriers, with every assumption labeled, and the ethical questions that arithmetic cannot settle

Nobody who works on bovine hemoglobin can avoid the question eventually. A skeptic at a conference, a journalist, a hospital administrator, or a patient's relative will ask it plainly: how much cattle does this take? It sounds like a joke and is not one. The question contains three separate inquiries, which are technical (how much hemoglobin is in a cow, and how much of it can be recovered), economic (what does the raw material cost and who controls it), and moral (is it right to use an animal this way, and does the answer change if the use saves human lives). This essay works through the first honestly, gestures at the second, and treats the third with the seriousness it deserves.

A warning about method. I am going to build an illustrative calculation from round numbers. The numbers are assumptions chosen to be plausible and are labeled as such, and none should be read as a measurement of any real manufacturing process. The purpose is to teach the shape of the reasoning, so that a reader can substitute better numbers and see how the answer moves.

The basic bookkeeping

Start with the animal. An adult beef animal weighs several hundred kilograms. Blood makes up a small percentage of body weight, and I will assume, for illustration, that a full-grown animal carries on the order of tens of liters of blood. I will also assume that cattle blood contains hemoglobin at a concentration broadly similar to human blood, in the neighborhood of a dozen grams per hundred milliliters. Real values vary with breed, age, health and the way blood is collected, and a reader with access to a veterinary physiology text should replace these with better ones.

Not all of that blood can be collected. At slaughter, a substantial fraction is retained in the carcass and organs, and only part flows out during bleeding. So the collectable volume per animal is less than the total, perhaps by a large factor. Then the blood must be processed: red cells separated from plasma, washed, lysed, the hemoglobin purified from the debris, cross-linked and polymerized, filtered and formulated. Each step loses material. A realistic yield, from collected hemoglobin to finished product, might be well under one hundred percent, and I will treat it here as an unknown fraction, call it y .

Now the demand side. A product dose is measured in grams of hemoglobin. The one bovine product with human approval in two countries has historically been supplied in bags of a couple of hundred milliliters at a concentration of roughly thirteen grams per deciliter, which is a few tens of grams per bag, according to the company's published descriptions. A patient who needs the product over a course of treatment may receive one bag or several. A veterinary dose is scaled to body weight and is typically a fraction of a bag.

Put the pieces together and the shape emerges. If one animal yields, after all losses, a quantity of finished hemoglobin equal to some number of bags, and a treatment course uses a few bags, then each animal supports a handful of treatments. That is the honest order of magnitude: not

one animal per patient, and not a thousand patients per animal, but somewhere in a range that depends on the yield and the dose, and that a reader can compute for themselves once the assumptions are fixed.

Where the number really lives

The calculation above is sensitive to two quantities more than any others. The first is the collection fraction. If blood is collected under conditions designed for the purpose, with anticoagulant, clean handling and controlled temperature, more can be recovered per animal than in ordinary slaughter, where blood is a byproduct of little value and is handled accordingly. The second is the process yield, which depends on the chemistry and on how much of the purified hemoglobin ends up in the target polymer size range. A process that discards the undesired fractions wastes more raw material than one that recycles them.

Both quantities are improvable, and both have been the subject of engineering effort. A manufacturer that doubles the yield halves the number of animals per patient. That is why the supply question is not a fixed feature of the technology. It is a function of process engineering, and the engineers have every incentive to improve it.

There is a further complication that a reader should keep in mind: the raw material must meet safety standards that ordinary slaughter blood does not. Earlier essays in this series discussed the sourcing regime for cattle, the herd controls and the regulatory limits set in response to prion disease. Those requirements restrict which animals can be used and therefore restrict supply independently of the arithmetic of hemoglobin content. A herd that meets the standard may be small, and a product that depends on it inherits the size.

A comparison that is easy to get wrong

The natural comparison is with blood donation. A donor gives a unit of whole blood at a session, and a unit of red cells contains a quantity of hemoglobin that is in the same general range as a few bags of the oxygen carrier. One might therefore conclude that an animal is a rough equivalent to a number of donors. That comparison is misleading in two ways.

First, a donor is renewable. A healthy adult can give again after a couple of months, and a population of donors supplies a continuous stream. An animal is used once. The comparison should therefore be between the donor population's annual output and the herd's annual turnover, not between one donor and one animal.

Second, the product and the donation are not equivalent clinically. A unit of red cells carries oxygen inside cells that last weeks and interact with the vasculature in the way nature designed. The carrier is a temporary bridge with a short circulating life. One does not simply exchange the units one for one. The relevant question is what the patient needs during the window in which blood is not available, and that is usually a much smaller quantity of hemoglobin than a full replacement.

The scale of the herd

To get a sense of proportion, consider the scale at which cattle are raised and slaughtered. Beef production worldwide is measured in many tens of millions of animals per year. In any single

country with a large beef industry, the annual slaughter runs into the millions. Compared with those figures, the number of animals that would be needed to supply an oxygen carrier for a specialized medical market is a very small fraction, unless the market becomes enormous.

I say this cautiously, because the fraction depends entirely on which market. A niche supply for trauma patients who cannot receive blood, and for veterinary practice, is one thing. A general substitute for transfusion across a health system would be another, and would require a supply chain on a different scale. The honest position is that the technology could supply the first from existing beef infrastructure, and that the second remains speculative until the product's clinical role is established. I do not have a defensible number for the second, and I would distrust anyone who offers one with confidence.

The byproduct argument

The strongest ethical argument in favor of bovine sourcing is that the blood is already there. Cattle are raised and slaughtered for food, and their blood is largely a waste stream or a low-value input to animal feed and industrial uses. If it can be converted into a medical product that saves human lives, the argument goes, no additional animal is killed on account of the product. The medical use adds value to something that would otherwise be discarded.

There is force in this. It converts what looks like an extra harm into a use of an existing one. Someone who accepts the existing practice of raising cattle for food can reasonably accept the use of the blood.

The argument is also incomplete. A byproduct that becomes valuable changes the economics of the primary product. If blood becomes worth a substantial amount per animal, it slightly raises the value of each animal and thereby slightly supports the industry that produces it. At small scale the effect is negligible. At large scale it is not. And the argument does nothing for a reader who objects to the industry itself. For that reader the question is whether a life-saving use is sufficient to justify participation in a practice they reject, and the byproduct framing does not answer it.

The objection from the other side

Some readers will object from a very different direction, not that using cattle is wrong, but that it is unnecessary, because human donors and, eventually, engineered sources could do the job. Recombinant hemoglobin, produced in bacteria or yeast, avoids the animal entirely. Hemoglobin from outdated human blood avoids the herd. Hemoglobin from other sources, such as marine worms, has been proposed.

These alternatives are real, and several are discussed elsewhere in this series. What can be said here is that each has its own difficulties. Recombinant production at the scale of grams per patient is expensive with current methods, and the molecules require engineering to overcome the same toxicities that plague other cell-free hemoglobins. Human-derived hemoglobin depends on the donor supply that the technology is meant to supplement. The bovine molecule has the advantage that it has been studied for decades, that its oxygen release does not depend on the cofactor that human hemoglobin needs, and that the raw material is available at scale. It is not the only possible answer. It is the answer with the most history.

What the animal cannot tell us

I want to say something about the moral status of the animal that is neither dismissive nor performatively grieving. Cattle are sentient, and the conditions in which they are raised and killed vary enormously. A responsible program would draw its blood from herds raised to high welfare standards, slaughtered by methods designed to minimize suffering, and audited by independent inspectors. Those requirements happen to align with the biosecurity requirements, since healthy, traceable animals are also the ones that meet the safety standards.

That alignment is fortunate, and it should not be mistaken for the whole of the ethical case. The deeper question is whether the interests of an animal can be outweighed by the interests of a person who would otherwise die. Most moral traditions answer yes in some form, and some answer no. The reader who says no is not being irrational, and a serious advocate of the technology should say so, and should respect the position rather than trying to talk the person out of it.

Religious and cultural considerations

There is a practical dimension of the ethical question that mainstream discussions rarely mention. Patients differ in their religious and cultural attitudes to animal-derived products. Some avoid products derived from particular species, and in a country where cattle are held sacred, the use of a bovine product in medicine may be contentious. In others, the objection may concern the manner of slaughter or the status of the blood itself.

Hospitals that serve diverse populations already deal with these concerns for other medicines, including gelatin capsules, heparin from pigs and heart valves from cattle and pigs. The established practice is disclosure and choice: patients are told the source and can decline. For a product intended for emergency use, when a patient may be unable to speak, the difficulty is sharper. The consent essays in this series discuss it. The relevant point here is that supply arithmetic and cultural acceptability interact. A product that a significant share of potential patients would refuse has a smaller effective market than the herd could supply.

The supply chain as a vulnerability

A last practical consideration is fragility. A product that depends on a specific herd, a specific plant, and a specific regulatory certification is exposed to shocks. A disease outbreak in the herd, a contamination event at the plant or a change in the regulation of cattle-derived material could interrupt supply for months. The history of the field includes companies whose entire operation rested on a single facility, and the closure of that facility ended the business.

A resilient design would diversify: several source herds in different regions, more than one manufacturing site, and a stockpile sufficient to bridge a temporary interruption. Each of those adds cost. In a low-margin market, the temptation is to skip them. The result is a technology that is theoretically abundant and practically brittle.

A worked example, with the assumptions exposed

To make the reasoning concrete, let me walk through a hypothetical, using invented round numbers that a reader can replace. Suppose that the blood collected from one animal, under

controlled conditions, contains hemoglobin equal to a hundred bags of finished product at the strength described earlier, before losses. Suppose that the overall yield from collected hemoglobin to finished, releasable product is one in three. Then one animal yields about thirty-three bags. If a typical course uses three bags, one animal supports about eleven courses.

Now vary the assumptions. If collection is half as efficient, the figure falls to five or six courses. If the yield doubles, it rises to twenty-odd. If a course uses only one bag, as in some veterinary cases and perhaps in some human bridge uses, the figure triples. The range from a handful to several dozen is not a scandal and not a miracle. It is a plain feature of the numbers.

The reader should keep two things in mind about this example. First, every input is hypothetical. I have not reported the actual yield of any manufacturing process, and a published figure from a company would supersede the assumptions above. Second, the interesting question is not the central estimate but the sensitivity. Whatever the true numbers are, the answer is likely to move by a factor of several with plausible changes in process, and therefore the honest response to the question in the title is a range with reasons.

What a decent answer sounds like

Suppose the skeptic at the conference asks the question again. A decent answer would run as follows. Each animal supports a small number of treatments, the exact number depends on process yield and dose, and I would rather show you the calculation than quote a figure. The blood comes from animals raised for food, which changes the moral picture without erasing it. The supply is adequate for a specialized market and untested for a general one. And the alternatives, recombinant and human-derived, are worth pursuing in parallel, because a field that depends on one source is fragile.

That answer contains no false precision and no evasion. It also leaves the moral question where it belongs, with the questioner. The arithmetic can tell you how many animals. It cannot tell you whether a life is worth them, and pretending otherwise is a way of avoiding the argument.

The measure that matters

There is a final reframing. The title asks how many cows is a life, and the question could be inverted. How many lives is a cow? If one animal supports, on the assumptions above, several patients whose alternative was death because compatible blood was unavailable, the ratio looks favorable to many people. If the patients would have survived anyway, or if the product does not reduce mortality, the ratio is meaningless.

That is why the clinical evidence matters more than the supply arithmetic. Supply is a constraint on scale, but benefit is what justifies the effort. A product that saves few lives per animal is still worth making if the lives are real and the alternative is nothing. A product that saves none is not worth a single animal. Everything else, including the yield calculations, is secondary to the question of whether the carrier works when it is needed. The essays elsewhere on this site examine that question, and this one has tried to make sure that the supply question is asked in the right order: after the evidence, not before it.

Questions worth putting to any manufacturer

A reader evaluating a company's claims about supply can put a short list of questions to it. How many animals are in the qualified herd, and how is that herd audited? What fraction of collected hemoglobin ends up in releasable product, and what happens to the rest? How many days of stock does the company hold, and what is the plan if the single plant goes offline? What is the cost of raw material per finished bag, and how much of the price is the herd and how much the process? Which independent body has inspected the welfare and biosecurity arrangements, and are its reports public?

None of these questions is hostile. A serious manufacturer will have answers, and the willingness to give them is itself informative. Vague answers about proprietary processes are a warning that the arithmetic has not been done, or has been done and is unflattering. In a field with a long record of promises that outran evidence, transparency about supply is one of the few things a patient community can reasonably demand.

Further reading

The manufacturing and sourcing of bovine-derived oxygen carriers is described in the published literature on the approved bovine product and in the company's own technical descriptions, which state the concentration and bag size assumed above. The regulatory framework for cattle-derived materials is set out in the United States Food and Drug Administration's rules on bovine spongiform encephalopathy, and comparable rules apply elsewhere. Beef production statistics are published by national agricultural agencies and by the United Nations Food and Agriculture Organization. All figures in the worked example are illustrative assumptions, not reported data.